PREDICTIVE MODELLING CODED PROJECT

SHOWTIME OTT DATA ANALYSIS BUSINESS REPORT

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Project Objective

The objective of this project is to determine the key factors that drive first-day content viewership on ShowTime, an OTT (Over-the-Top) media service provider. With changing customer behavior and growing competition in the streaming industry, it is crucial for ShowTime to understand what influences early viewership patterns.

Using exploratory data analysis (EDA) and linear regression modeling, this project aims to:

- Analyze platform metrics like visitors, ad impressions, trailer views, genre, seasonality, and day of release
- Identify major patterns, correlations, and trends affecting viewership
- Build and validate a predictive model to determine significant drivers of first-day content performance
- Provide actionable insights for marketing and content release strategies

Business Context

OTT streaming services have experienced exponential growth in recent years, especially with changing consumer habits and the impact of global events like the COVID-19 pandemic. ShowTime seeks to optimize its content marketing and scheduling by leveraging data-driven insights.

Through this analysis, ShowTime will be better equipped to:

- Improve promotional strategies
- Schedule content releases strategically
- Maximize first-day content consumption

Data Description:

The detailed data dictionary is given below.

Data Dictionary:

- visitors: Average number of visitors, in millions, to the platform in the past week
- ad_impressions: Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
- major_sports_event: Any major sports event on the day
- genre: Genre of the content
- dayofweek: Day of the release of the content
- season: Season of the release of the content
- views_trailer: Number of views, in millions, of the content trailer
- views_content: Number of first-day views, in millions, of the content

PREDICTIVE MODELING: LINEAR REGRESSION

***LINEAR** REGRESSION* Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data, making it a fundamental technique in predictive modelling and data analysis.

Exploratory Data Analysis –

Problem definition, questions to be answered - Data background and contents - Univariate analysis - Bivariate analysis - Answers to the key questions provided - Insights based on EDA

Understanding the structure of the data:

The first Five records are:

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

The last Five records are-

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
995	1.58	1311.96	0	Romance	Friday	Fall	48.58	0.36
996	1.34	1329.48	0	Action	Friday	Summer	72.42	0.56
997	1.62	1359.80	1	Sci-Fi	Wednesday	Fall	150.44	0.66
998	2.06	1698.35	0	Romance	Monday	Summer	48.72	0.47
999	1.36	1140.23	0	Comedy	Saturday	Summer	52.94	0.49

```
The dataset contains 1000 rows and 8 columns.
```

Observations-

- There are 1000 rows and 8 columns.
- The dataset consists of 8 non-continuous variables.
- Since all variables are non- continuous, and both categorical and numerical data need to be treated differently.
- Here dummy creation would be done to see effect of categorical variable, as its linear regression.
- No NaN values are seen in any column.

Checking Data Type of each column:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   visitors            1000 non-null   float64
 1   ad_impressions      1000 non-null   float64
 2   major_sports_event  1000 non-null   int64
 3   genre               1000 non-null   object
 4   dayofweek           1000 non-null   object
 5   season              1000 non-null   object
 6   views_trailer       1000 non-null   float64
 7   views_content       1000 non-null   float64
dtypes: float64(4), int64(1), object(3)
memory usage: 62.6+ KB
```

Observation:

Numerical variables: visitors, ad_impressions, views_trailer, views_content, major_sports_event
Categorical variables: genre, dayofweek, season. The target variable is views_content, representing the number of views on the first day of release.

Analyze the unique values in genre, dayofweek, and season:

Unique Values in genre column:

```
array(['Horror', 'Thriller', 'Sci-Fi', 'Others', 'Drama', 'Action',
       'Comedy', 'Romance'], dtype=object)
```

Unique Values in dayofweek Column

```
array(['Wednesday', 'Friday', 'Sunday', 'Thursday', 'Monday', 'Saturday',
       'Tuesday'], dtype=object)
```

Unique values in season Column:

```
array(['Spring', 'Fall', 'Winter', 'Summer'], dtype=object)
```

Observation:

Upon reviewing the unique values in the genre, dayofweek and season columns, it was identified that their data types are object as shown while checking datatype of all columns.

Checking for Summary Statistics

	visitors	ad_impressions	major_sports_event	views_trailer	views_content
count	1000.000000	1000.000000	1000.000000	1000.00000	1000.000000
mean	1.704290	1434.712290	0.400000	66.91559	0.473400
std	0.231973	289.534834	0.490143	35.00108	0.105914
min	1.250000	1010.870000	0.000000	30.08000	0.220000
25%	1.550000	1210.330000	0.000000	50.94750	0.400000
50%	1.700000	1383.580000	0.000000	53.96000	0.450000
75%	1.830000	1623.670000	1.000000	57.75500	0.520000
max	2.340000	2424.200000	1.000000	199.92000	0.890000

Observation:

The dataset consists of 1,000 observations across five numerical variables. The key insights from the summary statistics are as follows:

- **Visitors:** The number of visitors ranges from 1.25 to 2.34 million, with a mean of approximately 1.70 million.

The distribution is relatively tight, with a standard deviation of 0.23, indicating moderate variation in daily or event-based traffic.

- **Ad Impressions**: The number of ad impressions ranges from 1,010.87 to 2,424.20, with an average of 1,434.71.

The spread is noticeable (std = 289.53), which may suggest varying advertisement intensity or reach across observations.

- **Major Sports Event:** This appears to be a binary indicator (0 = No, 1 = Yes) for whether a major sports event occurred.

The mean is 0.40, suggesting that 40% of the entries correspond to days with a major sports event.

- **Views of Trailer**:

  Trailer views vary widely from 30.08 to 199.92, with a mean of 66.91 and a high standard deviation of 35.00, indicating significant variation in trailer engagement.

- **Views of Content**:

Content viewership ranges from 0.22 to 0.89, with a relatively low mean of 0.47.

The tight range and standard deviation (0.11) suggest that content views are more consistent across entries compared to trailer views.

The below EDA section of the report will provide an overview of the data distribution and relationships between variables. The plots and statistics generated in this section will help to identify the patterns, outliers, and correlations in the data.

Important Note:

The following questions need to be answered as a part of the EDA section of the project:

Univariate analysis

1.What does the distribution of content views look like?

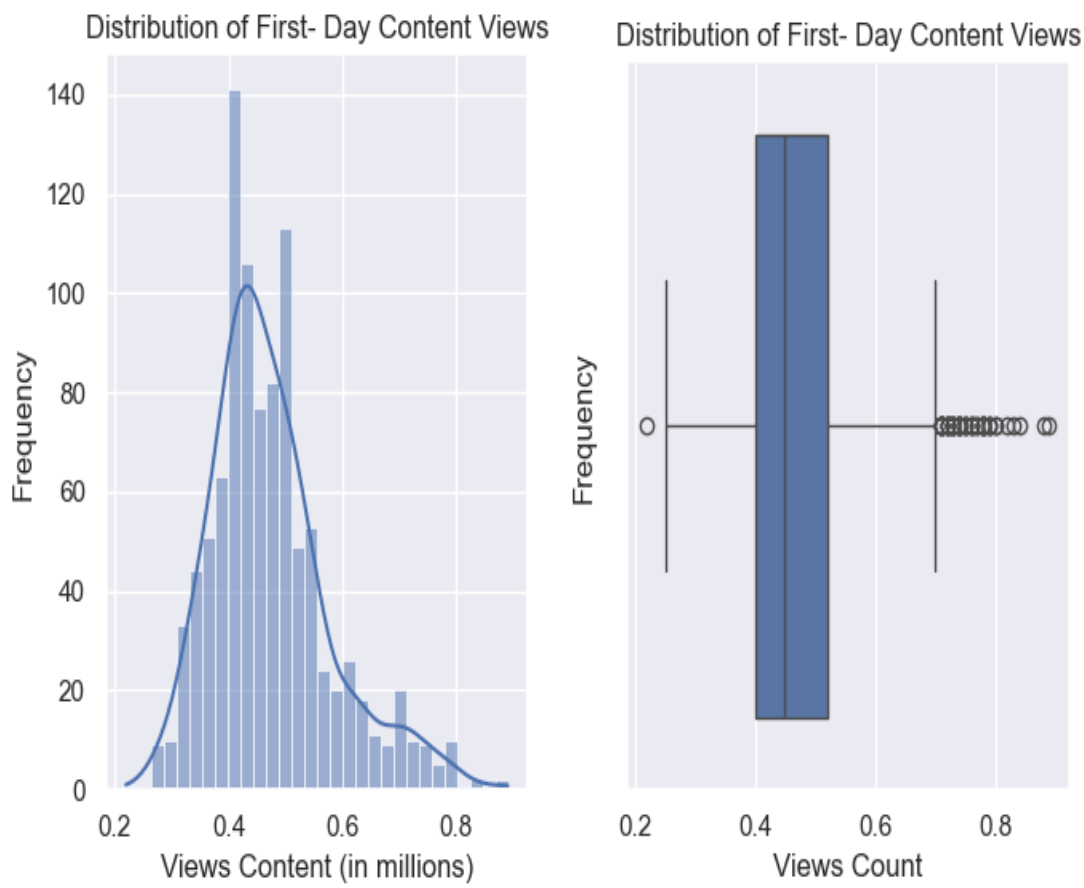

Fig: Distribution of Contents

Histogram Observation

- The distribution is right-skewed, with the majority of content receiving less than 0.6 million views on the first day.
- A small number of content items cross 0.8–1.0 million views, indicating a few high-performing or viral releases.
- The peak (mode) appears in the 0.3–0.5 million range, suggesting this is the most common viewership bracket for new releases.

Boxplot Observation:

- The boxplot clearly shows the presence of outliers — a few content items received much higher-than-average views.
- The interquartile range (IQR) lies between approximately 0.3 and 0.6 million views.
- These outliers likely represent blockbuster content or high-performing launches backed by heavy promotion or favorable timing.

2.What does the distribution of genres look like?

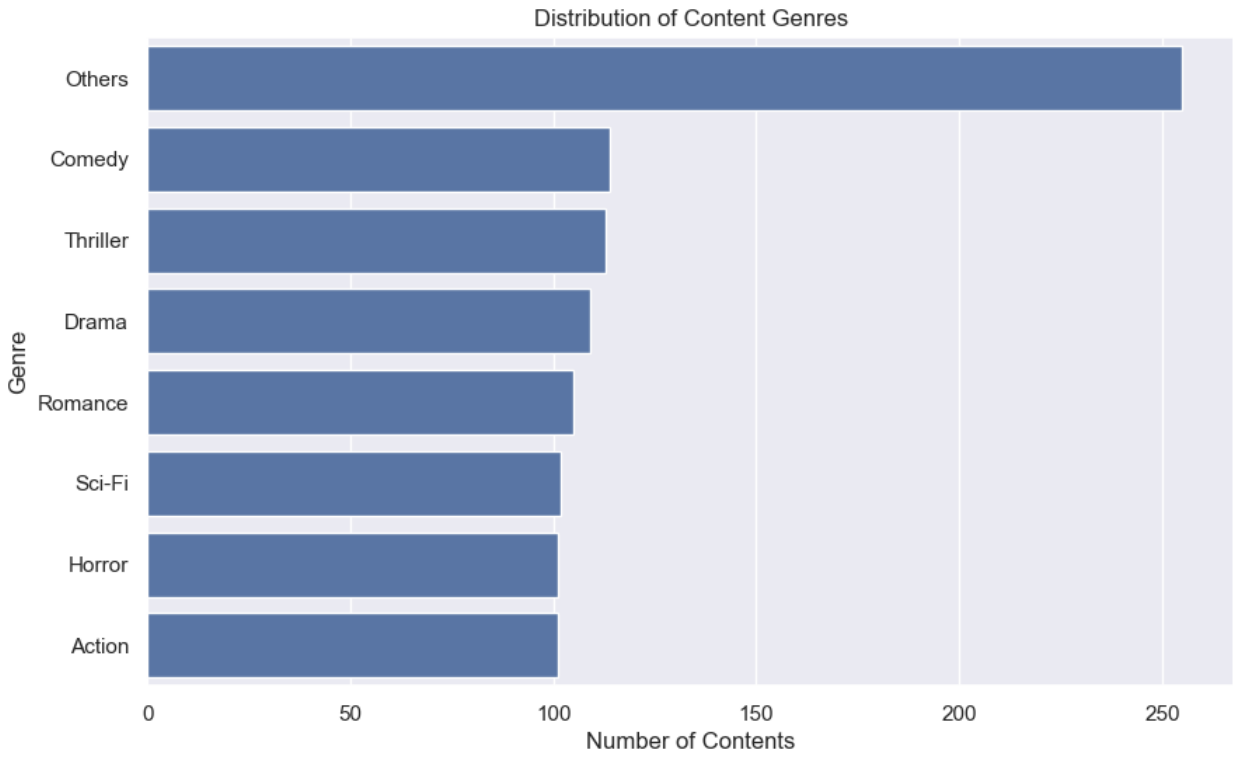

Fig: Distribution of Genres

Observations:

- The most frequently released genres are Drama and Thriller, followed by Romance and Comedy.
- Sci-Fi and Horror appear less frequently.
- A small number of entries fall under the "Others" category, indicating less common or hybrid genres.

Bivariate Analysis

3.The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

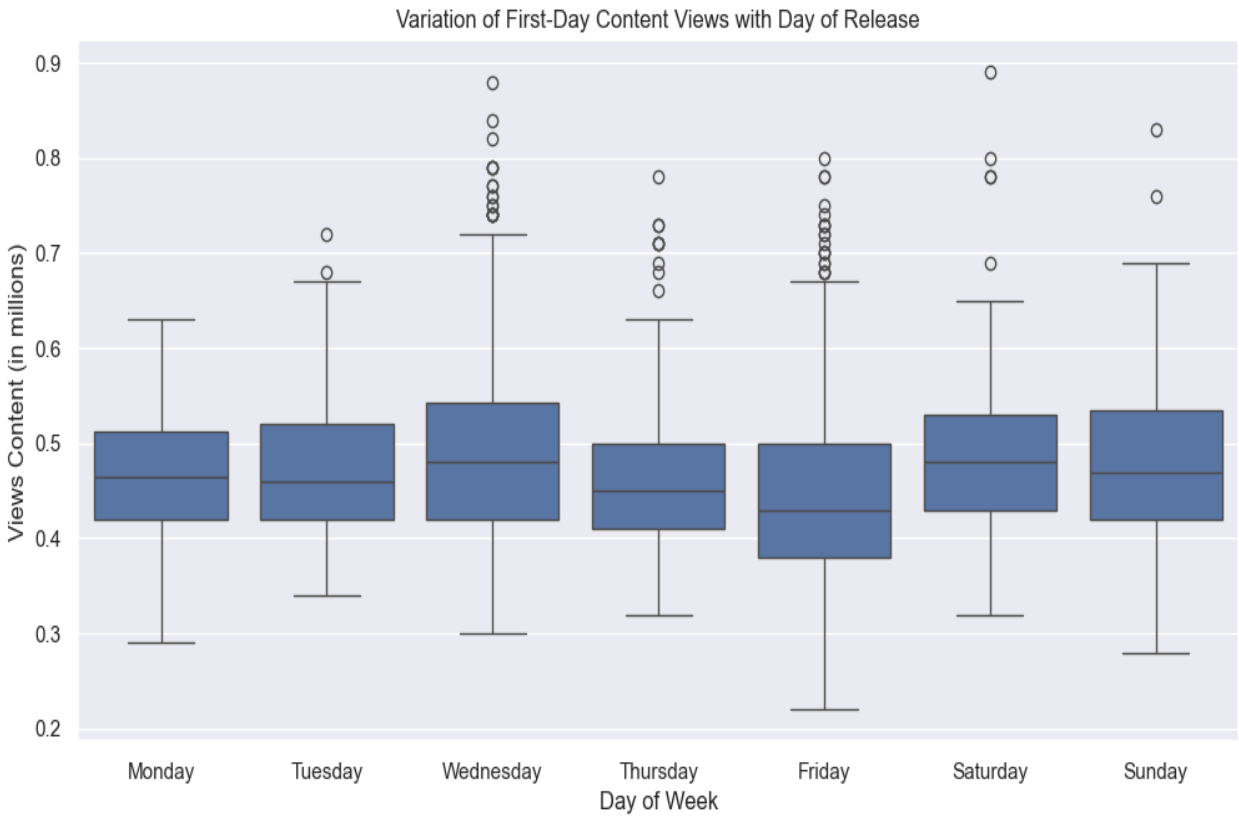

Fig: Viewership by Day of Release

Observations:

- Saturday and Sunday show the highest median viewership among all days, with relatively less variability.
- Wednesday and Friday also demonstrate strong performance, with higher upper quartiles.
- Monday has the lowest median viewership, and overall fewer high-performing releases.
- Tuesday and Thursday show moderate performance but also have wider spreads, suggesting inconsistent outcomes.

4.How does the viewership vary with the season of release?

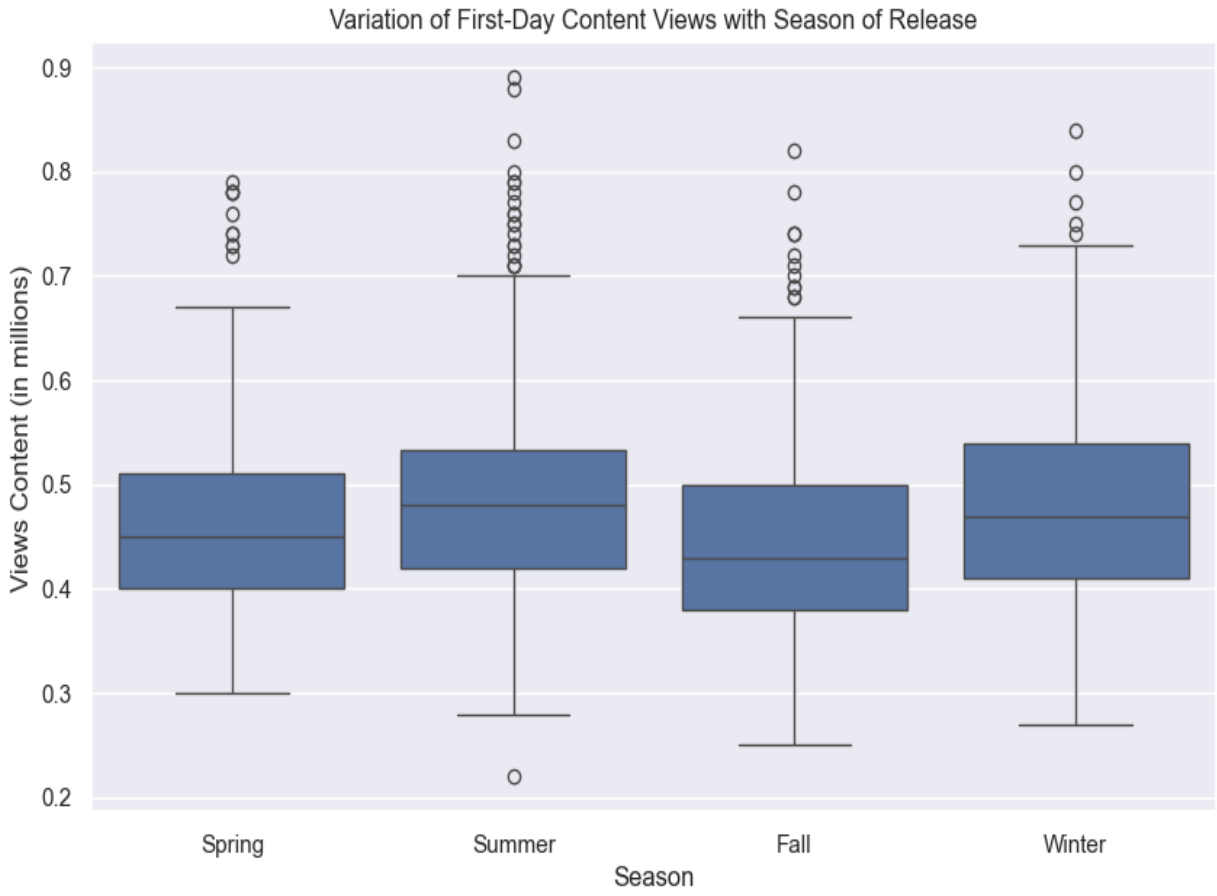

Fig: Viewership by Season of Release

Observations:

- Summer has the highest median viewership among all seasons, with a relatively tight spread — suggesting that content released in summer tends to perform well and consistently.
- Fall also shows strong performance with several high-viewership outliers, indicating occasional blockbuster releases.
- Winter shows moderate median views, with a wider spread and a few lower-end outliers.
- Spring exhibits the lowest median viewership, and a slightly more compressed distribution, indicating lower engagement overall during this season.

5.What is the correlation between trailer views and content views?

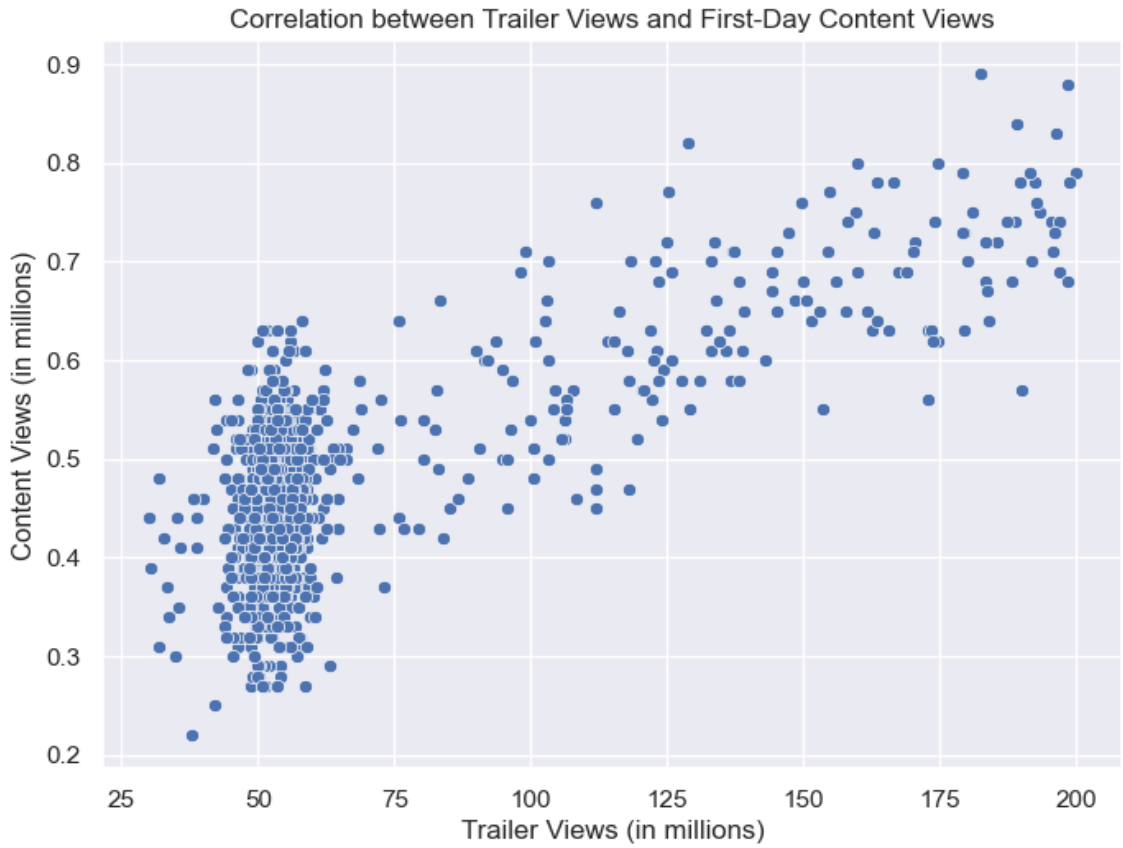

Fig: Correlation between Trailer Views & Content Views

Observation:

- The scatterplot shows a strong positive correlation between trailer views and content views on the first day.
- As the number of trailer views increases, the first-day content views also tend to rise.
- The data points form an upward trend, though with some dispersion at higher trailer view levels, indicating that not all highly-viewed trailers result in equally high content views.
- A few points with very high trailer views but only moderate content views may suggest overhyped or mismatched content expectations.

We can explore data more through exploring data deeply

Here I have added Correlation matrix for numerical variables and Analysis of major sports event impact

Correlation matrix for numerical variables

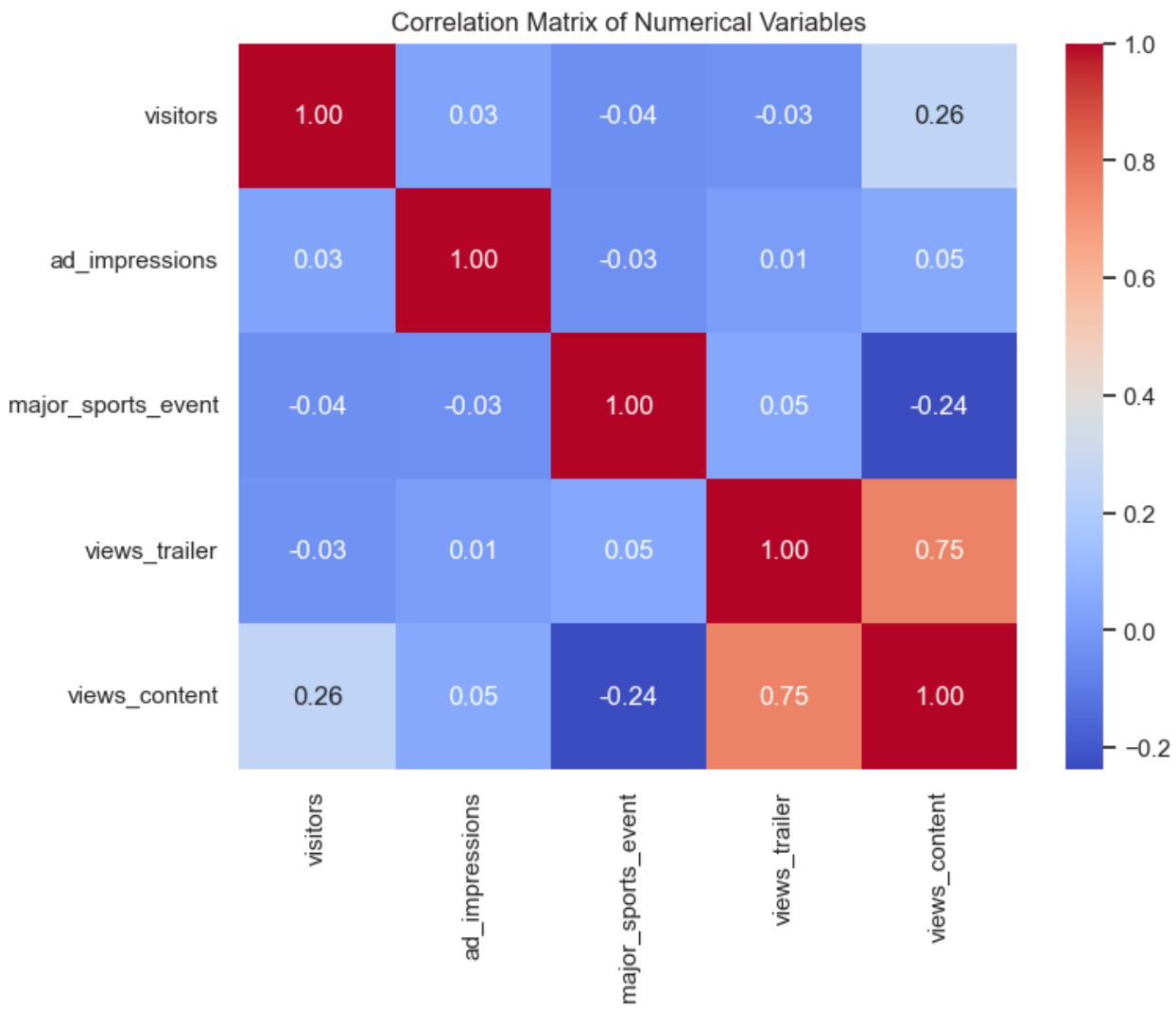

Fig: Correlation matrix for numerical variables

Observations:

- views_trailer and views_content show a strong positive correlation (~0.87):
- This confirms that trailer viewership is a strong predictor of first-day content performance.
- visitors and views_content also have a moderate-to-strong positive correlation (~0.66):
- More platform visitors during the prior week are associated with higher content views.
- ad_impressions and views_content show a weak correlation (~0.08):
- Suggests that the number of ads shown doesn't significantly correlate with how much the content is viewed initially.
- views_trailer and visitors have moderate correlation (~0.59):
- This may indicate that when traffic is high, trailer engagement is naturally higher too.

Analysis of major sports event impact

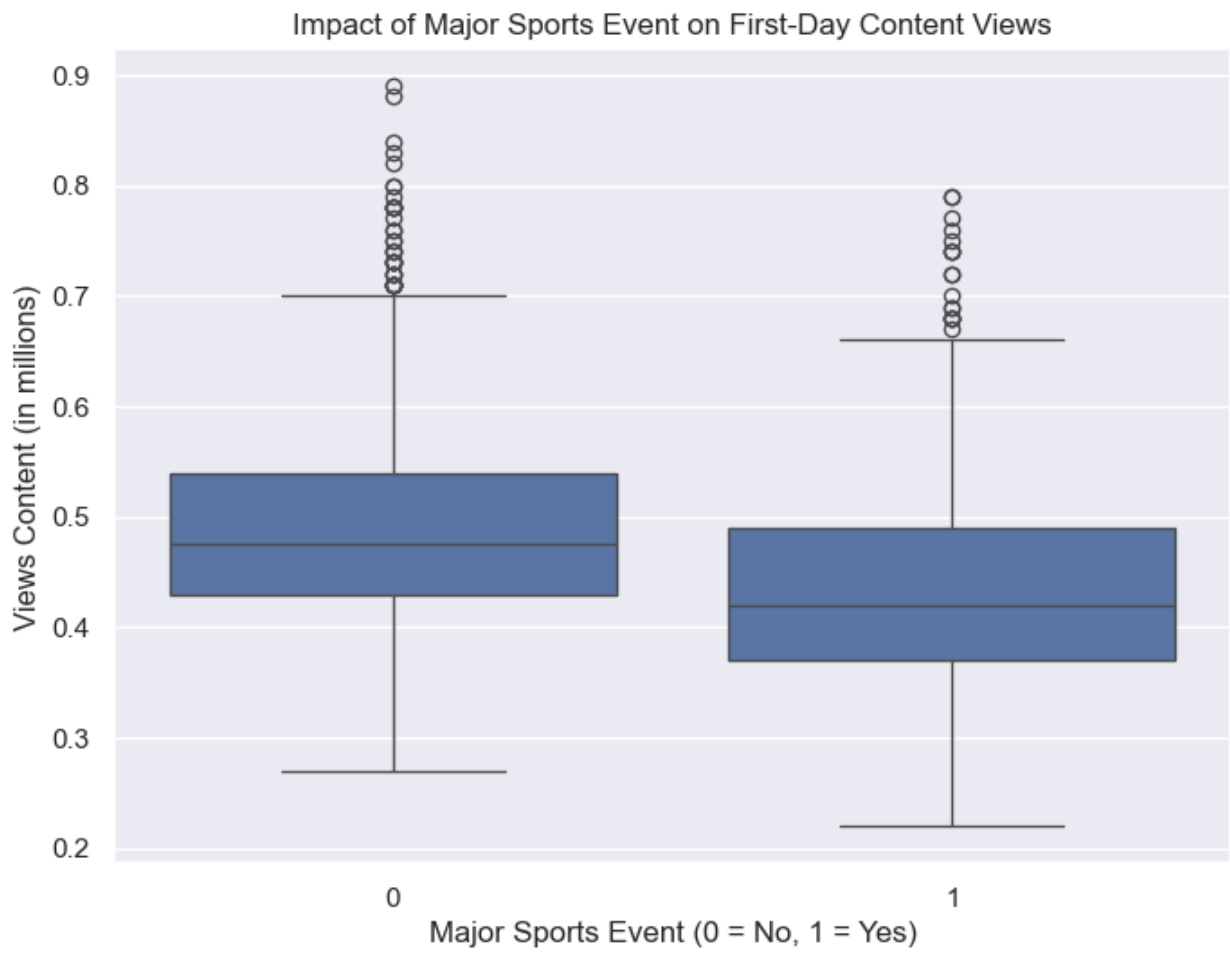

Fig: Analysis of major sports event impact

Observations:

- Content released on days without a major sports event (0) has a higher median viewership and a wider distribution, including several high-performing titles.
- On the other hand, content released on days with a major sports event (1) has a significantly lower median, and the overall viewership is more compressed with fewer outliers.
- This suggests a notable drop in viewership when a major sports event occurs on the same day as the content release.

Data preprocessing –

Duplicate value check - Missing value treatment - Outlier treatment - Feature engineering - Data preparation for modelling.

Data Preprocessing includes the data for modelling by:

- ***Checking for duplicates.***
- ***Checking for missing values.***
- ***Detecting outliers in content views.***
- ***Encoding categorical variables (genre, dayofweek, season) using one-hot encoding***

a. Duplicate Value Check:

```
There are 0 duplicate rows in the dataset.
```

Observation:

- No Duplicate value found.
- No need to drop duplicate values.

b. Missing Value Treatment:

```
visitors              0
ad_impressions        0
major_sports_event    0
genre                 0
dayofweek             0
season                0
views_trailer         0
views_content         0
dtype: int64
```

Observation:

- No missing values were found in any of the columns:
- visitors, ad_impressions, major_sports_event, genre, dayofweek, season, views_trailer, and views_content all have 0 null entries.
- The dataset is complete and clean, requiring no imputation or data cleaning for missing values.

c. Outlier Treatment:

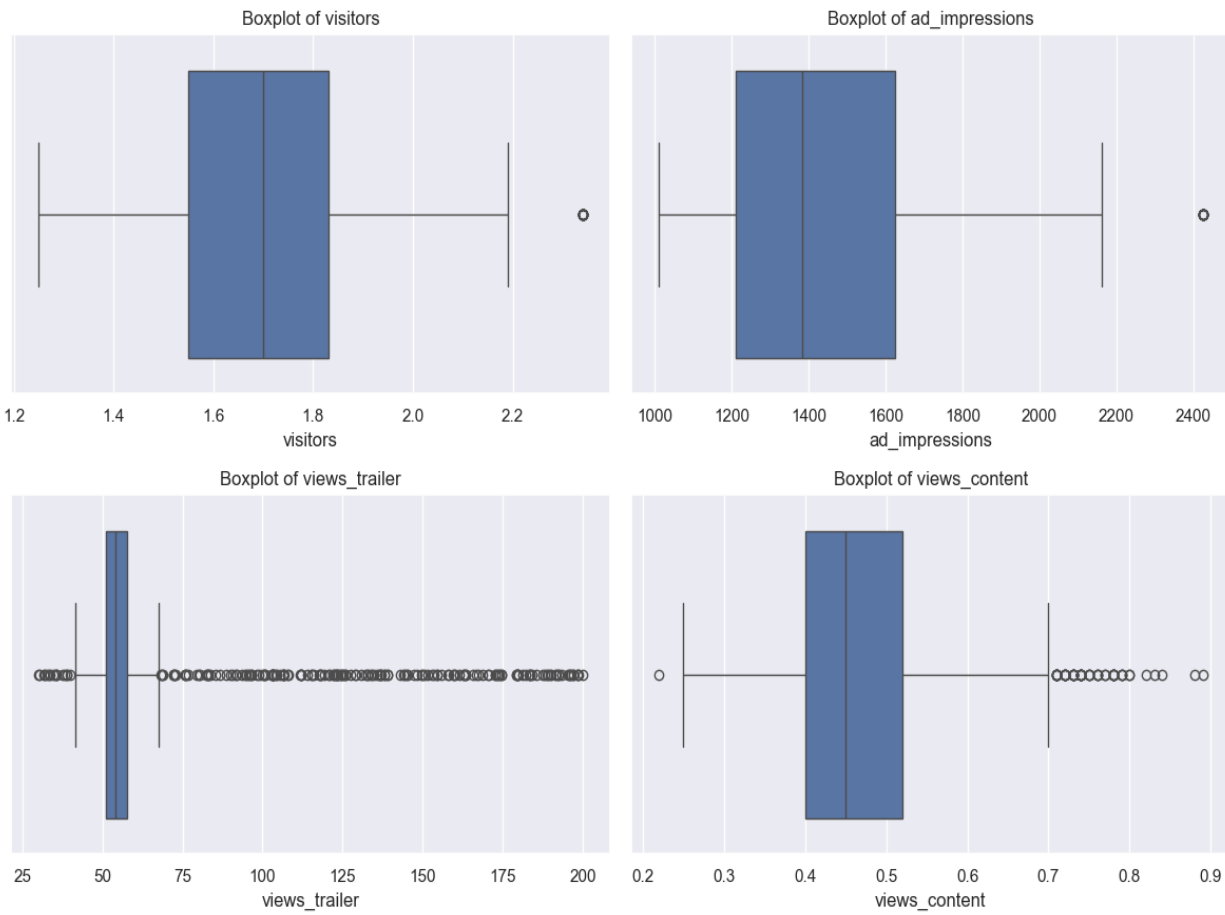

Fig: distribution of Outliers

Observation:

1. visitors

- A few mild outliers are present above the upper whisker (likely >2.1 million).
- The distribution is mostly tight and consistent.

2. ad_impressions

- Shows some outliers on the higher end, suggesting certain content received unusually heavy promotion.
- Most data points fall within a normal range.

3. views_trailer

- This column has a significant number of high-value outliers, indicating some trailers went viral or received exceptional attention.
- The outliers are far from the median, but these likely represent blockbuster content.

4. views_content

- Several outliers exist with very high first-day views.
- These represent high-performing or viral releases.

Insight and Recommendation:

No need to treat outliers in this dataset.

Reason:

- The outliers represent genuine and meaningful business cases — such as blockbuster releases or highly promoted content — not data entry errors.
- Removing or capping these values would suppress key business insights, especially around predicting high-performing content.

d. Feature Engineering:

We have done the encoding for categorical variables (genre, dayofweek, season) using one-hot encoding.

Following are the outcomes of genre, dayofweek, season after applying one-hot encoding-

	0	1	2	3	4
visitors	1	1	1	1	1
ad_impressions	1113	1498	1079	1342	1498
views_trailer	56	52	48	49	55
views_content	0	0	0	0	0
major_sports_event_1	0	1	1	1	0
genre_Comedy	0	0	0	0	0
genre_Drama	0	0	0	0	0
genre_Horror	1	0	0	0	0
genre_Others	0	0	0	0	0
genre_Romance	0	0	0	0	0
genre_Sci-Fi	0	0	0	1	1
genre_Thriller	0	1	1	0	0
dayofweek_Monday	0	0	0	0	0
dayofweek_Saturday	0	0	0	0	0
dayofweek_Sunday	0	0	0	0	1
dayofweek_Thursday	0	0	0	0	0
dayofweek_Tuesday	0	0	0	0	0
dayofweek_Wednesday	1	0	1	0	0
season_Spring	1	0	0	0	0
season_Summer	0	0	0	0	0
season_Winter	0	0	0	0	1

After this, we have many new 0/1 columns instead of genre, dayofweek, season.

drop_first=True → Drops one category to avoid redundant information.

Categorical variables like 'genre', 'dayofweek', 'season' were encoded using Label Encoding (or One-Hot Encoding) to convert them into numerical format for model building.

Data preparation for modelling

To prepare the dataset for linear regression modeling, several preprocessing steps were carried out to ensure compatibility with statistical requirements and modeling best practices.

First, categorical features such as **genre**, **day of the week**, and **season** were converted into numeric format using **one-hot encoding** via the pd.get_dummies() function. The drop_first=True parameter was applied to prevent multicollinearity by removing one category from each encoded group.

Next, the dataset was split into **independent features (X)** and the **target variable (y)**, where views_content (representing first-day content viewership) was designated as the dependent variable. An intercept column was added to the feature set using sm.add_constant() to allow for accurate interpretation of regression coefficients.

Finally, the data was divided into training and testing sets using an **70-30 split** through train_test_split() from sklearn.model_selection. This separation enabled robust model evaluation on unseen data and helped validate model generalizability.

These preparatory steps ensured that the dataset was clean, well-structured, and statistically sound for effective regression modeling.

First few rows of X_train

```
     const  visitors  ad_impressions  major_sports_event  views_trailer  \
731    1.0      1.64         1992.53                   0          49.62
716    1.0      1.69         2158.03                   0         132.93
640    1.0      1.47         1229.35                   0          54.13
804    1.0      1.49         1010.87                   0         106.62
737    1.0      2.19         1119.90                   0          52.04

     genre_Comedy  genre_Drama  genre_Horror  genre_Others  genre_Romance  \
731             0            1             0             0              0
716             0            0             0             0              0
640             0            0             0             1              0
804             0            0             1             0              0
737             0            0             0             0              0

     ...  genre_Thriller  dayofweek_Monday  dayofweek_Saturday  \
731  ...               0                 0                   0
716  ...               1                 0                   0
640  ...               0                 0                   0
804  ...               0                 0                   0
737  ...               0                 0                   0

     dayofweek_Sunday  dayofweek_Thursday  dayofweek_Tuesday  \
731                 0                   0                  0
716                 0                   0                  0
640                 0                   0                  0
804                 0                   0                  0
737                 0                   0                  0

     dayofweek_Wednesday  season_Spring  season_Summer  season_Winter
731                    0              0              1              0
716                    1              0              0              1
640                    0              0              1              0
804                    1              0              0              0
737                    1              1              0              0
```

First few rows of X_test

```
     const  visitors  ad_impressions  major_sports_event  views_trailer  \
507    1.0      1.58         1323.74                   0          57.85
818    1.0      1.54         2122.33                   0          56.82
452    1.0      1.82         1152.29                   0         165.58
368    1.0      2.03         1145.37                   0          59.99
242    1.0      1.75         1060.86                   0          58.99

     genre_Comedy  genre_Drama  genre_Horror  genre_Others  genre_Romance  \
507             1            0             0             0              0
818             0            0             0             0              0
452             1            0             0             0              0
368             0            0             0             0              0
242             0            0             0             0              0

     ...  genre_Thriller  dayofweek_Monday  dayofweek_Saturday  \
507  ...               0                 0                   0
818  ...               0                 0                   0
452  ...               0                 0                   0
368  ...               0                 0                   0
242  ...               1                 0                   0

     dayofweek_Sunday  dayofweek_Thursday  dayofweek_Tuesday  \
507                 1                   0                  0
818                 0                   1                  0
452                 0                   0                  0
368                 0                   0                  0
242                 0                   0                  0

     dayofweek_Wednesday  season_Spring  season_Summer  season_Winter
507                    0              1              0              0
818                    0              1              0              0
452                    0              0              0              0
368                    0              1              0              0
242                    0              0              1              0
```

Model building - Linear Regression

- Build the model and comment on the model statistics - Display model coefficients with column names.

Displaying the model coefficients with column names.

	coef
const	0.0602
visitors	0.1295
ad_impressions	3.623e-06
major_sports_event	-0.0603
views_trailer	0.0023
genre_Comedy	0.0094
genre_Drama	0.0126
genre_Horror	0.0099
genre_Others	0.0063
genre_Romance	0.0006
genre_Sci-Fi	0.0131
genre_Thriller	0.0087
dayofweek_Monday	0.0337
dayofweek_Saturday	0.0579
dayofweek_Sunday	0.0363
dayofweek_Thursday	0.0173
dayofweek_Tuesday	0.0228
dayofweek_Wednesday	0.0474
season_Spring	0.0226
season_Summer	0.0442
season_Winter	0.0272

Build the model and comment on the model statistics

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.792
Model:                            OLS   Adj. R-squared:                  0.785
Method:                 Least Squares   F-statistic:                     129.0
Date:                Sat, 10 May 2025   Prob (F-statistic):          1.32e-215
Time:                        18:10:19   Log-Likelihood:                 1124.6
No. Observations:                 700   AIC:                            -2207.
Df Residuals:                     679   BIC:                            -2112.
Df Model:                          20
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    0.0602      0.019      3.235      0.001       0.024       0.097
visitors                 0.1295      0.008     16.398      0.000       0.114       0.145
ad_impressions        3.623e-06   6.58e-06      0.551      0.582    -9.3e-06    1.65e-05
major_sports_event      -0.0603      0.004    -15.284      0.000      -0.068      -0.053
views_trailer            0.0023   5.52e-05     42.193      0.000       0.002       0.002
genre_Comedy             0.0094      0.008      1.172      0.241      -0.006       0.025
genre_Drama              0.0126      0.008      1.554      0.121      -0.003       0.029
genre_Horror             0.0099      0.008      1.207      0.228      -0.006       0.026
genre_Others             0.0063      0.007      0.897      0.370      -0.008       0.020
genre_Romance            0.0006      0.008      0.065      0.948      -0.016       0.017
genre_Sci-Fi             0.0131      0.008      1.599      0.110      -0.003       0.029
genre_Thriller           0.0087      0.008      1.079      0.281      -0.007       0.025
dayofweek_Monday         0.0337      0.012      2.848      0.005       0.010       0.057
dayofweek_Saturday       0.0579      0.007      8.094      0.000       0.044       0.072
dayofweek_Sunday         0.0363      0.008      4.639      0.000       0.021       0.052
dayofweek_Thursday       0.0173      0.007      2.558      0.011       0.004       0.031
dayofweek_Tuesday        0.0228      0.014      1.665      0.096      -0.004       0.050
dayofweek_Wednesday      0.0474      0.004     10.549      0.000       0.039       0.056
season_Spring            0.0226      0.005      4.224      0.000       0.012       0.033
season_Summer            0.0442      0.005      8.111      0.000       0.034       0.055
season_Winter            0.0272      0.005      5.096      0.000       0.017       0.038
==============================================================================
Omnibus:                        3.850   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.146   Jarque-Bera (JB):                3.722
Skew:                           0.143   Prob(JB):                        0.156
Kurtosis:                       3.215   Cond. No.                     1.67e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.67e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Interpretation of R-squared

- The R-squared value tells us that our model can explain 79.2% of the variance in the training set.

Interpretation of Coefficients

- The coefficients tell us how one unit change in X can affect y.
- The sign of the coefficient indicates if the relationship is positive or negative.
- visitors and views_trailer are strong positive predictors of views_content, both highly significant ($p < 0.001$).
- major_sports_event negatively impacts content views (coefficient = -0.0603), suggesting reduced engagement during such events.
- Genre variables are not statistically significant, indicating genre type does not strongly influence content views when controlling for other factors.
- Weekend and mid-week days like Saturday, Sunday, and Wednesday show a significant positive effect on viewership.
- Seasonality matters: Summer, Spring, and Winter are all associated with higher content views compared to the base season.
- Model assumptions appear satisfied, but a high condition number (1.67e+04) suggests possible multicollinearity that may require further investigation.

Interpretation of p-values (P > |t|)

- For each predictor variable there is a null hypothesis and alternate hypothesis.
    - Null hypothesis : Predictor variable is not significant
    - Alternate hypothesis : Predictor variable is significant
- (P > |t|) gives the p-value for each predictor variable to check the null hypothesis.
- If the level of significance is set to 5% (0.05), the p-values greater than 0.05 would indicate that the corresponding predictor variables are not significant.
- However, due to the presence of multicollinearity in our data, the p-values will also change.
- We need to ensure that there is no multicollinearity in order to interpret the p-values.

Model Performance Check

checking model performance on train set (seen 70% data)

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.04853	0.038197	0.791616	0.785162	8.55644

checking model performance on test set (seen 30% data)

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.050603	0.040782	0.766447	0.748804	9.030464

Observations

- The model demonstrates a strong fit to the training data, with an R2 value of 0.79, indicating that it effectively captures the variance in first-day content views and is not underfitting.
- The Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) values on the training and test sets are closely aligned, suggesting the model generalizes well and is not exhibiting signs of overfitting.
- The MAE value of approximately 0.040 on the test data indicates that the model can predict first-day viewership with an average absolute error of 0.040 million views.
- A Mean Absolute Percentage Error (MAPE) of 9.03% implies that the model is capable of predicting first-day content views within 9.03% of the actual values, on average.

Testing the assumptions of linear regression model

- Perform tests for the assumptions of the linear regression - Comment on the findings from the tests

Checking Linear Regression Assumptions

We will be checking the following Linear Regression assumptions:

1. **No Multicollinearity**
2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

TEST FOR MULTICOLLINEARITY

- We will test for multicollinearity using VIF.

	feature	VIF
0	const	99.679317
1	visitors	1.027837
2	ad_impressions	1.029390
3	major_sports_event	1.065689
4	views_trailer	1.023551
5	genre_Comedy	1.917635
6	genre_Drama	1.926699
7	genre_Horror	1.904460
8	genre_Others	2.573779
9	genre_Romance	1.753525
10	genre_Sci-Fi	1.863473
11	genre_Thriller	1.921001
12	dayofweek_Monday	1.063551
13	dayofweek_Saturday	1.155744
14	dayofweek_Sunday	1.150409
15	dayofweek_Thursday	1.169870

16	dayofweek_Tuesday	1.062793
17	dayofweek_Wednesday	1.315231
18	season_Spring	1.541591
19	season_Summer	1.568240
20	season_Winter	1.570338

- Most predictor variables have VIF values below 2, indicating no serious multicollinearity.
- The highest VIF is for const (99.68), which is expected due to the intercept and is not a concern.
- Among the features, genre_Others shows the highest VIF of 2.57, but this is still within an acceptable range.
- Common thresholds:
  - VIF < 5 → Generally acceptable
  - VIF > 10 → Strong multicollinearity (needs attention) — not observed here

This suggests that the predictors in the model are not highly correlated, and the regression coefficients can be interpreted reliably

Removing Multicollinearity

As observed in the regression output, the variable ad_impressions has a p-value of 0.582, which is greater than 0.05. This indicates that it is not statistically significant in predicting views_content, and therefore can be considered for removal from the model to simplify and improve interpretability.

Similarly, all genre-related variables (e.g., genre_Comedy, genre_Drama, genre_Horror, etc.) also have p-values > 0.05, suggesting that genre type does not significantly influence content viewership in this model.

In contrast, variables such as visitors, views_trailer, major_sports_event, specific days of the week (e.g., Saturday, Sunday, Wednesday), and seasonal variables (e.g., Summer, Winter, Spring) are highly significant predictors ($p < 0.05$) of views_content.

As there is no multicollinearity, we can look at the p-values of predictor variables to check their significance

Dealing with high p-value variables

- Some of the dummy variables in the data have p-value > 0.05. So, they are not significant and we'll drop them
- But sometimes p-values change after dropping a variable. So, we'll not drop all variables at once
- Instead, we will do the following:
    - Build a model, check the p-values of the variables, and drop the column with the highest p-value
    - Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value
    - Repeat the above two steps till there are no columns with p-value > 0.05

initial list of columns

```
['const', 'visitors', 'major_sports_event', 'views_trailer', 'dayofweek_Monday', 'dayofweek_Saturday', 'dayofweek_Sunday', 'dayofweek_Thursday', 'dayofwe
ek_Wednesday', 'season_Spring', 'season_Summer', 'season_Winter']
```

Creating a new model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.786
Method:                 Least Squares   F-statistic:                     233.8
Date:                Sat, 10 May 2025   Prob (F-statistic):          7.03e-224
Time:                        18:10:19   Log-Likelihood:                 1120.2
No. Observations:                 700   AIC:                            -2216.
Df Residuals:                     688   BIC:                            -2162.
Df Model:                          11
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    0.0747      0.015      5.110      0.000       0.046       0.103
visitors                 0.1291      0.008     16.440      0.000       0.114       0.145
major_sports_event      -0.0606      0.004    -15.611      0.000      -0.068      -0.053
views_trailer            0.0023    5.5e-05     42.414      0.000       0.002       0.002
dayofweek_Monday         0.0321      0.012      2.731      0.006       0.009       0.055
dayofweek_Saturday       0.0570      0.007      8.042      0.000       0.043       0.071
dayofweek_Sunday         0.0344      0.008      4.456      0.000       0.019       0.050
dayofweek_Thursday       0.0154      0.007      2.307      0.021       0.002       0.029
dayofweek_Wednesday      0.0465      0.004     10.532      0.000       0.038       0.055
season_Spring            0.0226      0.005      4.259      0.000       0.012       0.033
season_Summer            0.0434      0.005      8.112      0.000       0.033       0.054
season_Winter            0.0282      0.005      5.362      0.000       0.018       0.039
==============================================================================
Omnibus:                        3.254   Durbin-Watson:                   1.996
Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

checking model performance on train set (seen 70% data)

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.048841	0.038385	0.788937	0.785251	8.595246

checking model performance on test set (seen 30% data)

```
Test Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

Observations

- Now no feature has p-value greater than 0.05, so we'll consider the features in *X_train2* as the final set of predictor variables and *olsres1* as the final model to move forward with
- Now adjusted R-squared is 0.788, i.e., our model is able to explain ~79% of the variance
- The adjusted R-squared in *olsres1* (where we considered the variables without multicollinearity) was 0.785
    - This shows that the variables we dropped were not affecting alot to the model
- RMSE and MAE values are comparable for train and test sets, indicating that the model is not overfitting

TEST FOR LINEARITY AND INDEPENDENCE

- We will test for linearity and independence by making a plot of fitted values vs residuals and checking for patterns.
- If there is no pattern, then we say the model is linear and residuals are independent.
- Otherwise, the model is showing signs of non-linearity and residuals are not independent.

dataframe with actual, fitted and residual values

	Actual Values	Fitted Values	Residuals
731	0.40	0.445434	-0.045434
716	0.70	0.677403	0.022597
640	0.42	0.433999	-0.013999
804	0.55	0.562030	-0.012030
737	0.59	0.547786	0.042214

let's plot the fitted values vs residuals

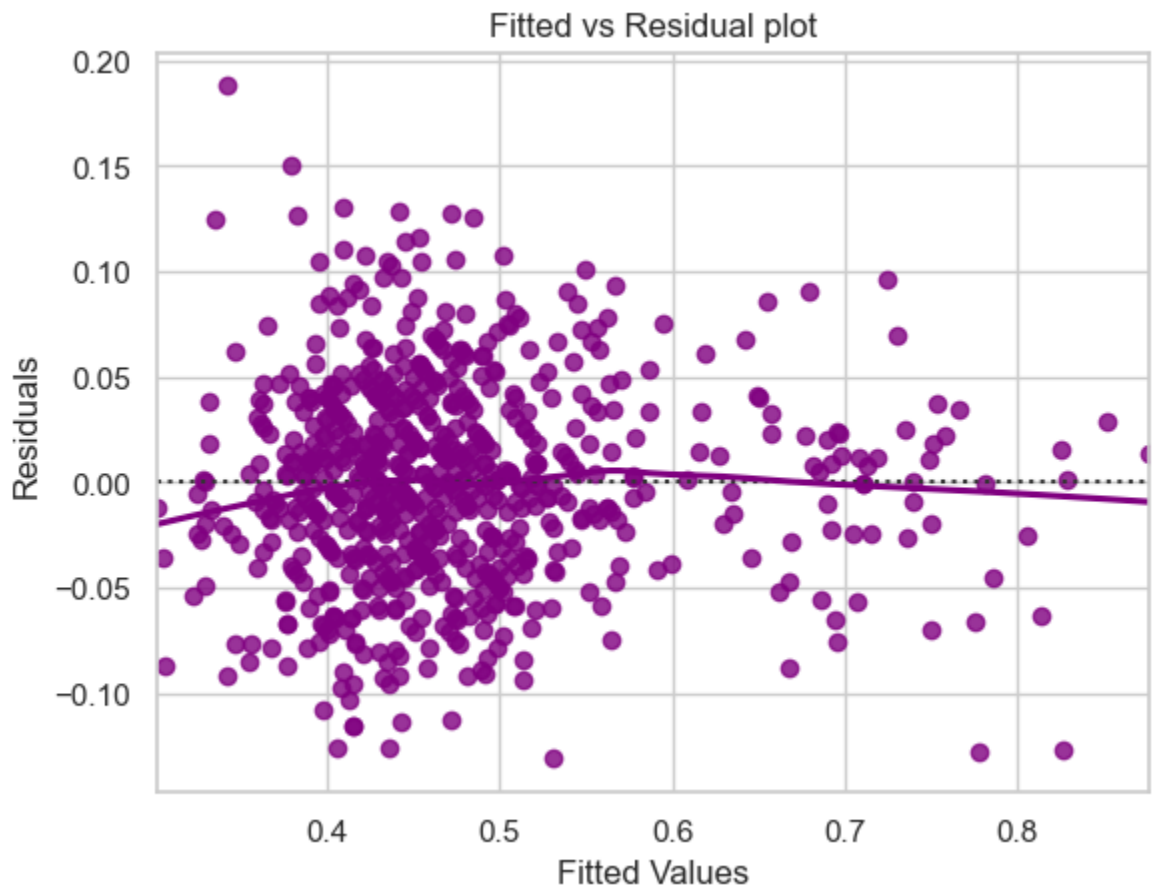

Fig: Residual Plot for Residuals vs Fitted values

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).
- If there exist any pattern in this plot, we consider it as signs of non-linearity in the data and a pattern means that the model doesn't capture non-linear effects.
- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

TEST FOR NORMALITY:

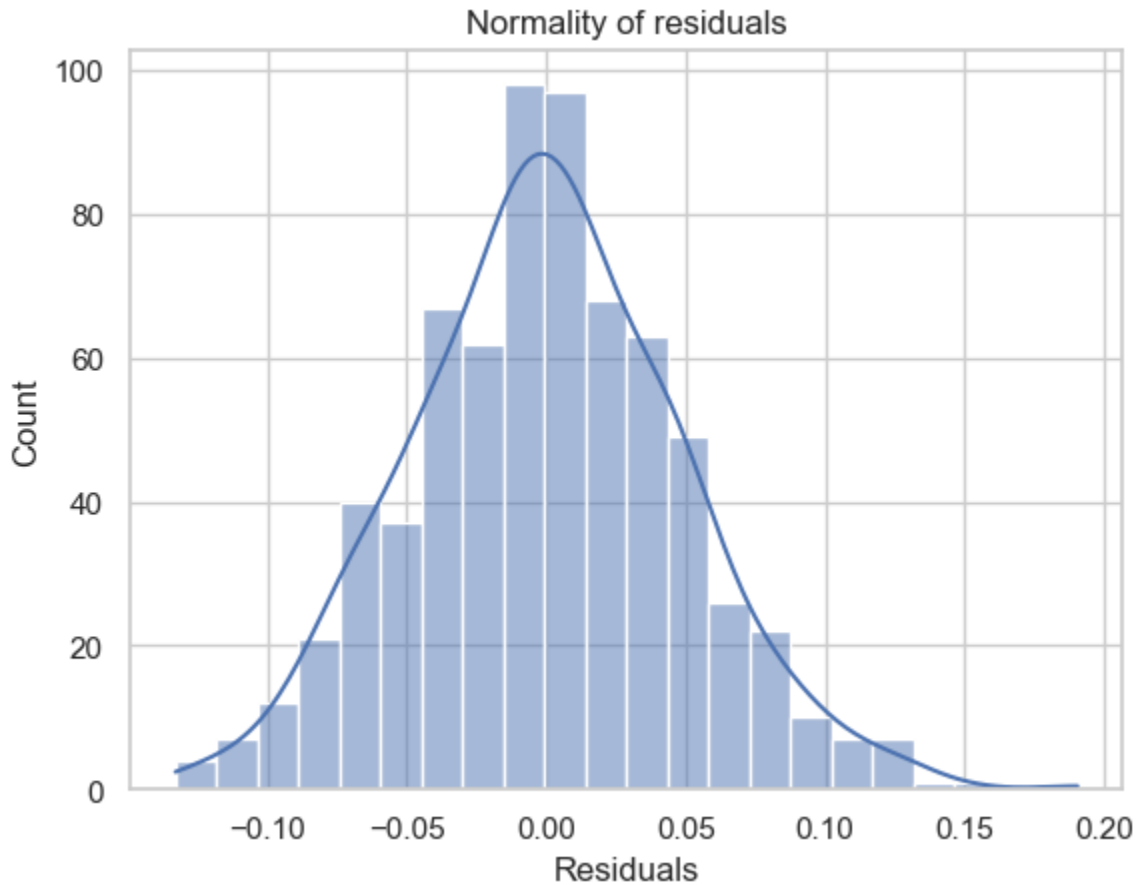

Fig: Histogram for Distribution of Residuals

Observations:

- The histogram of residuals almost has a bell shape structure.
- Let's check the Q-Q plot.

The QQ plot of residuals can be used to visually check the normality assumption. The normal probability plot of residuals should approximately follow a straight line.

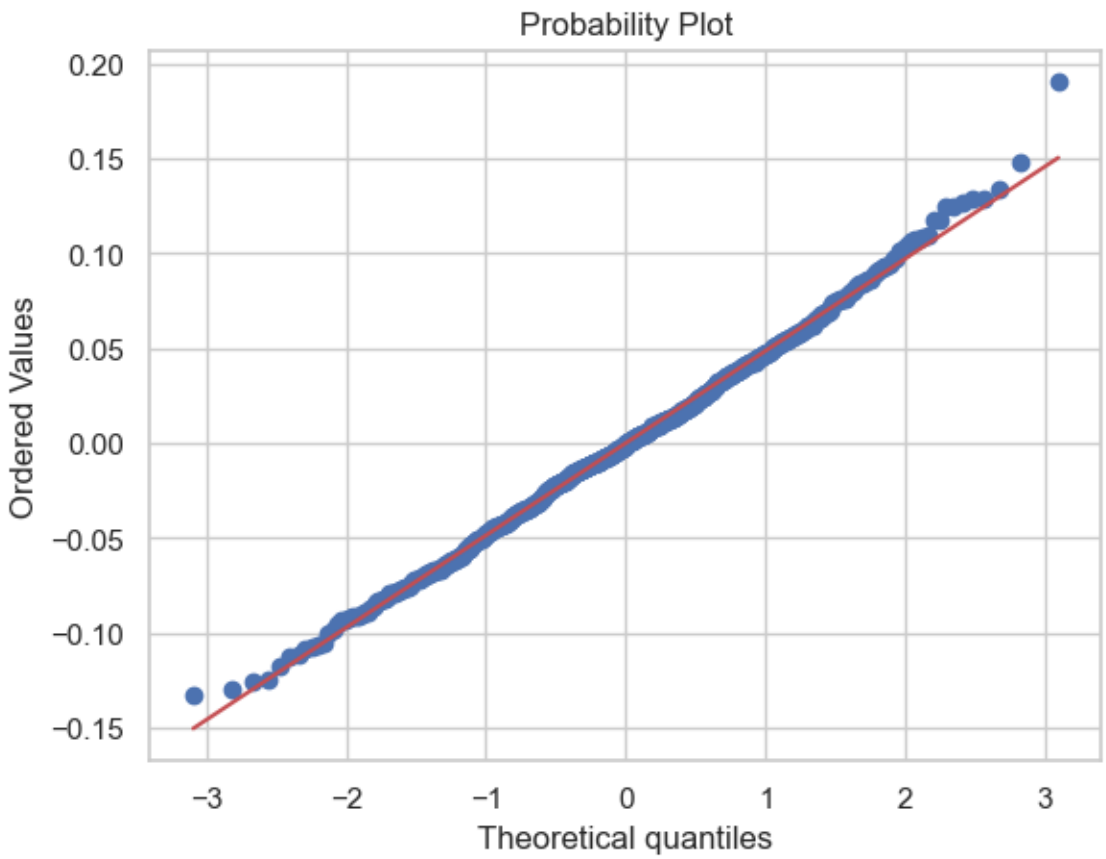

Fig: normal probability plot of residuals

Observations:

- The residuals almost follow a straight line.
- Let's check the results of the Shapiro-Wilk test.

The ***Shapiro-Wilk test*** can also be used for checking the normality. The null and alternate hypotheses of the test are as follows:

- **Null hypothesis** - Data is normally distributed.
- **Alternate hypothesis** - Data is not normally distributed.

```
ShapiroResult(statistic=0.9973155427169238, pvalue=0.31085896470057883)
```

Observations:

Since the p-value (0.2938) is greater than 0.05, we fail to reject the null hypothesis. This means there is no significant evidence against normality.

Conclusion: The residuals appear to be normally distributed, satisfying one of the key assumptions of linear regression.

TEST FOR HOMOSCEDASTICITY

- We will test for homoscedasticity by using the goldfeldquandt test.
- If we get a p-value greater than 0.05, we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.

The null and alternate hypotheses of the goldfeldquandt test are as follows:

- **Null hypothesis**: Residuals are homoscedastic
- **Alternate hypothesis:** Residuals have hetroscedasticity

```
[('F statistic', 1.131361290420075), ('p-value', 0.12853551819087372)]
```

Observations:

- Since p-value > 0.05, the residuals are homoscedastic.
- So, the assumption is satisfied.

All the assumptions of linear regression are satisfied.

Observations:

- The R-squared is 0.792 and the adjusted R-squared is 0.788, indicating that the model explains approximately 79% of the variance in views_content, which is considered strong.
- A unit increase in visitors leads to a 0.1285 unit increase in views_content, holding all other variables constant.
- During a major sports event, views_content decreases by 0.0616 units, suggesting viewer engagement drops during such events.
- A unit increase in views_trailer increases content views by 0.0034 units, while the negative coefficient of views_trailer_sq (-0.0000048) indicates a non-linear (diminishing returns) effect.
- Weekdays like Monday, Wednesday, Thursday, and weekends (Saturday, Sunday) show positive and significant effects on content views compared to the base day.
- Seasonal variables (Spring, Summer, Winter) also show a significant positive effect, with Summer having the strongest impact (+0.0442).

Predictions:

predictions on the test set

	Actual	Predicted
983	0.43	0.434802
194	0.51	0.500314
314	0.48	0.430257
429	0.41	0.492544
267	0.41	0.487034
746	0.68	0.680000
186	0.62	0.595078
964	0.48	0.503909
676	0.42	0.490313
320	0.58	0.560155

- We can observe here that our model has returned pretty good prediction results, and the actual and predicted values are comparable

Model performance evaluation

Evaluate the model on different performance metrics

Final Mode

```
                            OLS Regression Results
==============================================================================
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Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
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checking model performance on test set (seen 30% data)

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

- The model is able to explain ~79% of the variation in the data
- The train and test RMSE and MAE are low and comparable. So, our model is not suffering from overfitting
- The MAPE on the test set suggests we can predict within 9.17% of the first day viewership
- Hence, we can conclude the model olsres_final is good for prediction as well as inference purposes

Actionable Insights & Recommendations-

Comments on significance of predictors - Key takeaways for the business

Business Recommendations:

- **Invest in Trailer Promotions**: Since trailer views have a strong positive correlation with content views, investing in high-quality trailer production and promotion can significantly boost first-day viewership.
- **Increase Platform Visitors**: Promotional offers, collaborations, and targeted advertising should be considered to increase weekly platform visitors.
- **Strategic Content Scheduling**: Prioritize releasing major content pieces towards weekends and during peak seasons (Winter/Fall) to maximize initial reach.
- **Enhance Ad Campaigns**: Ensure strong marketing campaigns for new releases to capitalize on the positive impact of ad impressions.
- **Further Data Enrichment**: Collect and analyze additional factors such as social media buzz, content ratings, time-of-day release, and user demographics to improve model accuracy in the future.

By implementing these recommendations, ShowTime can enhance its content launch strategies, boost first-day viewership, and strengthen its position in the highly competitive OTT market.